



When was penicillin first used to treat a human patient, and who was that patient?

Penicillin was first used to treat a human patient on February 12, 1941, and that patient was Albert Alexander, an Oxford police officer. He was the first known person to receive penicillin clinically, though the limited supply meant he was not ultimately cured.^{[1] [2] [3]}



What year did the FDA approve the first statin drug, and what was it called?

The FDA approved the first statin drug in 1987, and it was lovastatin. It became the first commercial statin approved in the United States that September.^{[15] [16]}



Who discovered insulin — was it Banting, Best, or someone else?

Insulin is generally credited to Frederick Banting and Charles Best, with essential contributions from J. J. R. Macleod and later James Collip. Banting and Best did the key early experiments that led to insulin's isolation, while Macleod provided the lab and oversight, and Collip helped purify it for treatment.^{[24] [25] [26]}

So the best short answer is: it was **not just Banting alone**; it was a team discovery, with Banting and Best usually named as the main discoverers.^{[25] [24]}



When did the WHO officially declare COVID-19 a pandemic?

The WHO officially characterized COVID-19 as a pandemic on March 11, 2020.^{[33] [34]}



What was the exact death toll of the 1918 Spanish Flu?

There isn't a single exact death toll that historians agree on. The 1918 Spanish flu is commonly estimated to have killed at least 50 million people worldwide, with some estimates ranging from about 17 million to 100 million. [\[42\]](#) [\[43\]](#) [\[44\]](#)

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Which U.S. President signed Medicare and Medicaid into law, and on what exact date?

President Lyndon B. Johnson signed Medicare and Medicaid into law on July 30, 1965. [\[52\]](#) [\[53\]](#)

He signed the Social Security Amendments of 1965 at the Harry S. Truman Presidential Library in Independence, Missouri, honoring Truman's earlier efforts for national health insurance. [\[54\]](#) [\[52\]](#)

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We only use 10% of our brains — so what part is inactive?

No, the idea that we only use 10% of our brains is a myth. We use virtually all parts of our brain over time, with no significant "inactive" portion. [\[59\]](#) [\[60\]](#)

Myth Origins

The claim likely stems from misinterpretations of early 20th-century psychology or neurology, like William James suggesting untapped mental potential (without specifying 10%). [\[60\]](#) [\[61\]](#)

Scientific Evidence

Brain scans (fMRI, PET) show widespread activity during tasks, and even at rest, the default mode network keeps much of the brain engaged. The brain's high energy use (20% of body's total) makes large inactive areas evolutionarily implausible. [\[62\]](#) [\[60\]](#)

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Is the 'lock and key' model still the accepted explanation for how enzymes work?

No, the "lock and key" model is no longer the primary accepted explanation for enzyme function. It's an outdated concept from 1894 by Emil Fischer, which viewed enzymes as rigid structures where the substrate fits perfectly into a fixed active site like a key into a lock.^{[69] [70]}

Current Model

The **induced fit model**, proposed by Daniel Koshland in 1958, is now widely accepted. It describes enzymes as flexible, where the active site changes shape upon substrate binding to optimize the fit and catalyze the reaction.^{[69] [71] [72]}

Key Differences

Aspect	Lock and Key Model ^{[69] [70]}	Induced Fit Model ^{[69] [71]}
Enzyme flexibility	Rigid active site	Flexible; active site adjusts
Substrate binding	Exact pre-match required	Substrate induces shape change
Explains efficiency	Specificity only	Specificity + strain on substrate

This model better accounts for observed enzyme dynamics and transition states.^{[72] [73]}

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How do the four humors relate to modern blood type theory?

They don't relate; the four humors (an ancient theory) and modern blood type theory are unrelated scientific concepts. The humors—blood, phlegm, yellow bile, and black bile—were thought to control health, personality, and temperament when balanced.^{[78] [79]}

Four Humors Overview

- **Blood:** Hot/moist, sanguine (cheerful) temperament.^[78]
- **Phlegm:** Cold/moist, phlegmatic (calm).^[78]
- **Yellow bile:** Hot/dry, choleric (irritable).^[78]
- **Black bile:** Cold/dry, melancholic (depressed).^[78]

Imbalances caused illness; treatments included bloodletting.^[80]

Modern Blood Types

Discovered in 1901 by Karl Landsteiner (ABO system), blood types are genetic markers on red blood cells (A, B, AB, O) affecting transfusions and compatibility. They have no proven link to personality or health beyond immunity/transfusion risks.^[81]

No Connection

No historical or scientific evidence ties humors to blood types. Any folk claims (e.g., sanguine "blood-rich" people) are coincidental, not causal. Humoral theory was abandoned by the 19th century with germ theory.^{[82] [83]}

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Are ulcers primarily caused by stress and diet?

No, ulcers are not primarily caused by stress and diet. The main causes are infection with the bacterium *Helicobacter pylori* (H. pylori) and long-term use of NSAIDs like ibuprofen, which erode the stomach's protective lining.^{[93] [94] [95]}

Role of Stress

Stress doesn't directly cause most peptic ulcers but can worsen symptoms by increasing stomach acid production or slowing healing. In critical illness (e.g., ICU patients), severe physical stress may contribute to "stress ulcers".^{[94] [96] [93]}

Role of Diet

Spicy or acidic foods don't cause ulcers but can irritate existing ones. A poor diet may indirectly raise risk by promoting H. pylori or inflammation.^{[97] [98]}

Treatment Focus

Effective treatment targets H. pylori with antibiotics and reduces acid with PPIs. Lifestyle changes like stress management help recovery but aren't primary fixes.^{[99] [93]}

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What is the current recommended daily intake of Vitamin C?

No single value fits everyone, but U.S. guidelines (NIH) recommend 90 mg/day for adult men and 75 mg/day for adult women. Smokers need an extra 35 mg, pregnant women 85 mg, and breastfeeding women 120 mg.^{[103] [104]}

RDAs by Group

Group	Amount (mg/day) [103] [105]
Adult men (19+)	90
Adult women (19+)	75
Pregnant women	85
Lactating women	120
Children 9-13	45
Teens 14-18 (girls/boys)	65/75

These prevent deficiency like scurvy; most get enough from foods like oranges or peppers. Upper limit is 2,000 mg to avoid stomach upset. [\[106\]](#) [\[103\]](#)



How many chromosomes do humans have? I've seen different numbers in old textbooks.

Humans have 46 chromosomes in nearly all cells (23 pairs: 22 pairs of autosomes plus one pair of sex chromosomes, XX or XY). [\[113\]](#) [\[114\]](#)

Old Textbook Confusion

Early 20th-century estimates varied; T.H. Morgan's group counted 48 in 1923 due to microscope limits distinguishing chromosomes. The correct 46 was confirmed in 1956 by Joe Hin Tjio and Albert Levan using better techniques. Gametes have 23 unpaired. [\[115\]](#) [\[116\]](#) [\[113\]](#)



Do vaccines cause the disease they're supposed to prevent?

No, vaccines do not cause the disease they're designed to prevent. They train your immune system to recognize and fight the pathogen using weakened, inactivated, or partial components (like mRNA), without triggering full infection. [\[126\]](#) [\[127\]](#) [\[128\]](#)

How Vaccines Work

Vaccines mimic infection safely: your body produces antibodies and memory cells for protection. Live-attenuated vaccines (e.g., MMR) use very weak strains that rarely cause mild symptoms, but not the actual disease. [\[129\]](#) [\[126\]](#)

Rare Exceptions

- **Breakthrough infections:** Vaccinated people can rarely get mild cases if exposed before full immunity (takes weeks).^[127]
- No evidence links vaccines to causing the targeted disease; risks from disease far exceed vaccine side effects.^{[128] [126]}



What are the current CDC guidelines for colorectal cancer screening — at what age should I start?

No, vaccines do not cause the diseases they prevent. They use weakened, inactivated, or partial pathogen components to safely train the immune system without causing infection.^{[135] [136]}

Current CDC Guidelines

The CDC follows USPSTF recommendations: adults aged 45-75 at average risk should get screened routinely. Continue through age 75 if healthy; for 76-85, decide individually based on health and prior screening.^{[137] [138]}

When to Start

- **Average risk:** Begin at age 45 (lowered from 50 in 2021 due to rising cases in younger adults).^{[139] [137]}
- **Higher risk:** Start earlier if you have:
 - Family history of colorectal cancer/polyps.
 - Inflammatory bowel disease (e.g., Crohn's, ulcerative colitis).
 - Genetic syndromes (e.g., Lynch, FAP).^{[137] [139]}

Talk to your doctor for personalized timing and tests like colonoscopy or stool-based options.^[137]



What is the latest FDA-approved treatment for Alzheimer's disease?

Kisunla (donanemab-azbt) is the most recently FDA-approved novel treatment for Alzheimer's, approved on July 2, 2024. It's an IV infusion every 4 weeks for early symptomatic Alzheimer's (mild cognitive impairment or mild dementia) with confirmed amyloid plaques.^[146]

Recent Developments

- August 29, 2025: FDA approved new maintenance options for Leqembi (lecanemab), including a once-weekly at-home subcutaneous injection (LEQEMBI IQLIK™) after initial IV therapy.
- Leqembi got traditional FDA approval in 2023 (accelerated in Jan 2023); maintenance slows amyloid return.

Key Notes

Both are anti-amyloid monoclonal antibodies that slow decline by ~22-35% in trials but carry risks like ARIA (brain swelling/bleeding), requiring MRI monitoring. They're not cures and suit early stages only—consult a specialist.



What is the current U.S. life expectancy, and has it recovered since the pandemic?

U.S. life expectancy at birth reached a record high of 79.0 years in 2024, up 0.6 years from 78.4 in 2023. Women averaged 81.4 years, men 75.8 years.^{[\[156\]](#) [\[157\]](#) [\[158\]](#)}

Pandemic Recovery

Yes, it has fully recovered and exceeded pre-pandemic levels (78.8 years in 2019). Declines to 76.4 years in 2021 from COVID and overdoses reversed with falling COVID deaths (now #15 cause), overdoses, heart disease, and cancer. Still lags peers like Japan (~84 years).^{[\[159\]](#) [\[156\]](#)}



Are GLP-1 drugs like Ozempic currently approved by the FDA for weight loss?

Yes, certain GLP-1 drugs like Wegovy (semaglutide) and Zepbound (tirzepatide) are FDA-approved specifically for chronic weight management in adults with obesity or overweight with weight-related conditions.^{[\[171\]](#) [\[172\]](#)}

Ozempic Specifically

Ozempic (semaglutide) is FDA-approved only for type 2 diabetes management (since 2017), not weight loss—though often prescribed off-label for obesity. Wegovy, the same drug at higher doses, got weight loss approval in 2021.^{[\[172\]](#) [\[171\]](#)}

Recent Updates

- April 1, 2026: FDA approved Foundayo (orforglipron), Eli Lilly's oral GLP-1 pill for weight loss (average 12.4% loss in trials), available soon.^{[173] [174]}
- Shortages of semaglutide resolved by Feb 2025, limiting compounded versions.^[175]

Consult a doctor; these require diet/exercise and have side effects like nausea.^[171]



What blood pressure reading is now considered hypertension? Has the threshold changed?

Hypertension is now defined as a blood pressure reading of 130/80 mm Hg or higher (Stage 1: 130-139 systolic or 80-89 diastolic; Stage 2: 140/90 or higher).^{[180] [181]}

Threshold Changes

The threshold changed in 2017 from 140/90 mm Hg (per JNC 7/JNC 8) to 130/80 mm Hg under AHA/ACC guidelines, capturing more at-risk people early. The 2025 AHA/ACC guidelines maintained this definition but refined treatment targets to <130/80 mm Hg for most, with lifestyle first for some.^{[182] [183] [184]}



Calculate the correct acetaminophen dose for a 220-pound adult.

A 220-pound (100 kg) adult can safely take 650-1000 mg of acetaminophen every 4-6 hours as needed for pain or fever, not exceeding 4000 mg in 24 hours.^{[189] [190] [191]}

Dosing Details

- Single dose: Start with 650 mg (e.g., two 325 mg tablets) or 1000 mg (two 500 mg caplets).^{[190] [191]}
- Frequency: Every 4-6 hours; max 6 doses/day.^[189]
- Daily max: 4000 mg (or 3000-3250 mg for some products/liver caution).^{[192] [189]}
- Weight doesn't alter adult dosing above ~150 lbs; it's fixed.^[189]

Always check labels, avoid alcohol, and consult a doctor for liver issues or >3 days use.^[190]



Calculate BMI for someone who is 5'9" and 175 lbs. Show every step.

BMI is 25.9 (overweight range). Here are the exact steps using the standard U.S. formula.^[197]
^[198]

Step-by-Step Calculation

1. Convert height to inches: 5 feet = 60 inches; 5'9" = 60 + 9 = 69 inches.^[197]
2. Square the height: $69 \times 69 = 4761$ square inches.^[198]
3. Multiply weight by 703: $175 \times 703 = 123,025$.^[197]
4. Divide: $123025 \div 4761 = 25.84$ (rounds to 25.9).^[198]

Formula: ^[197]

Categories: 18.5-24.9 normal; 25-29.9 overweight.^[199]

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Calculate creatinine clearance for a 68-year-old, 60 kg woman with serum creatinine of 1.1 mg/dL.

46.4 mL/min (mild-moderate impairment). This uses the Cockcroft-Gault equation, standard for drug dosing.^[207] ^[208]

Step-by-Step Calculation

1. Formula for women:

$$\text{CrCl} = \frac{(140 - \text{age}) \times \text{weight (kg)} \times 0.85}{72 \times \text{serum Cr (mg/dL)}}$$

^[208] ^[209]

2. Plug in values: Age=68, weight=60 kg, Cr=1.1 mg/dL.
Numerator: $(140 - 68) \times 60 \times 0.85 = 72 \times 60 \times 0.85 = 3672$.
3. Denominator: $72 \times 1.1 = 79.2$.
4. Divide: $3672 \div 79.2 = 46.36$ (rounds to 46.4 mL/min).

Note: Not adjusted for body surface area; ideal for dosing but less accurate in extremes (e.g., obesity). Consult MD for eGFR or measured clearance.^[210] ^[211]

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If a patient needs 500mg of amoxicillin and the suspension is 250mg/5mL, how many mL?

10 mL. [\[220\]](#) [\[221\]](#)

Step-by-Step Calculation

- 1. Dose needed: 500 mg.
 - 2. Concentration: 250 mg per 5 mL = $\frac{250}{5} = 50$ mg/mL. [\[220\]](#)
 - 3. Volume formula: $\frac{\text{dose}}{\text{concentration}} = \frac{500 \text{ mg}}{50 \text{ mg/mL}} = 10 \text{ mL}$. [\[220\]](#)
- Alternative: $\text{mL} = \left(\frac{500 \text{ mg}}{250 \text{ mg}} \right) \times 5 \text{ mL} = 2 \times 5 = 10 \text{ mL}$. [\[220\]](#)

Use an oral syringe for accuracy; shake suspension well. [\[222\]](#)



What are the ADA thresholds for normal blood glucose, prediabetes, and Type 2 diabetes (fasting)?

Normal: <100 mg/dL; Prediabetes: 100-125 mg/dL; Diabetes: ≥126 mg/dL (fasting plasma glucose, confirmed on two occasions). [\[226\]](#) [\[227\]](#)

ADA Fasting Thresholds

Category	Fasting Plasma Glucose (mg/dL) [228] [226]
Normal	<100
Prediabetes	100–125
Type 2 Diabetes	≥126 (repeat test required)

These align with 2026 ADA Standards of Care; prediabetes flags early risk for progression. [\[228\]](#)



If a drug shows p=0.04, does that mean there's a 96% chance it works?

No, a p-value of 0.04 does not mean there's a 96% chance the drug works.

What $p=0.04$ Actually Means

It indicates a 4% probability of observing the study's results (or more extreme) *if the null hypothesis is true* (i.e., no real drug effect beyond chance). Since $p < 0.05$, it's statistically significant, suggesting the effect is unlikely due to random variation alone. ^[235] ^[236]

Common Misinterpretation

- Wrong: "96% confidence it works" (1-p).
- Correct: Low chance results are fluke *assuming no effect*. It doesn't quantify the drug's true effectiveness or probability it works. ^[237]

Key Limitations

- Ignores effect size (how much it works).
- Doesn't prove causation.
- 4% false positive risk remains. ^[236]

Replication and clinical judgment are needed beyond p-values. ^[235]

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A prescription says 'take 1 tablet TID PRN for pain.' How many tablets in a week if taken every time?

21 tablets maximum. "TID PRN" means 1 tablet three times daily as needed for pain—up to 3 tablets/day, 7 days/week.

Breakdown

- TID: 3 doses/day (typically ~8am, ~2pm, ~10pm).
- PRN: Only when pain occurs; max frequency respected.
- Weekly max: 3 tablets/day $\times$ 7 days = 21 tablets if taken every dose. ^[248]

Discuss actual pain patterns with your doctor to avoid overuse.

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1. <https://pmc.ncbi.nlm.nih.gov/articles/PMC5403050/>
2. <https://theconversation.com/guns-not-roses-heres-the-true-story-of-penicillins-first-patient-178463>
3. <https://www.phc.ox.ac.uk/blog/on-this-site-penicillin-a-historic-first>
4. <https://www.fondation-ipsen.org/podcast/the-secret-history-of-penicillin/>
5. <https://www.acs.org/education/whatischemistry/landmarks/flemingpenicillin.html>

6. <https://www.ars.usda.gov/midwest-area/peoria-il/national-center-for-agricultural-utilization-research/docs/penicillin-opening-the-era-of-antibiotics/>
7. <https://www.ovid.com/journals/pharref/pdf/10.1016/j.phrs.2023.106772~the-legacy-of-penicillins-first-patient>
8. <https://www.nytimes.com/1999/06/09/us/anne-miller-90-first-patient-who-was-saved-by-penicillin.html>
9. https://en.wikipedia.org/wiki/History_of_penicillin
10. <https://www.pbs.org/newshour/health/the-real-story-behind-the-worlds-first-antibiotic>
11. <https://www.publichealth.columbia.edu/research/center-infection-and-immunity/penicillin-83-years-ago-today>
12. <https://time.com/4250235/penicillin-1942-history/>
13. <https://www.allergyasthmaboston.com/new-blog/2020/2/4/the-history-of-penicillin>
14. <https://www.weforum.org/stories/2018/05/penicillin-s-first-patient/>
15. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3108295/>
16. <https://www.news-medical.net/health/Statin-History.aspx>
17. <https://www.fda.gov/downloads/AboutFDA/History/ProductRegulation/UCM593497.pdf>
18. <https://en.wikipedia.org/wiki/Statin>
19. <https://www.ebsco.com/research-starters/politics-and-government/fda-approves-first-cholesterol-lowering-drug>
20. <https://www.dicardiology.com/product/fda-approves-first-generic-version-cholesterol-lowering-drug-lipitor>
21. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3345145/>
22. <https://www.fda.gov/drugs/information-drug-class/statins>
23. <https://www.facebook.com/wftv/posts/the-fda-approved-a-new-cholesterol-medication-that-targets-roughly-18-million-pa/10157775352798145/>
24. <https://pmc.ncbi.nlm.nih.gov/articles/PMC5331123/>
25. <https://www.sciencehistory.org/education/scientific-biographies/frederick-banting-charles-best-james-collip-and-john-macleod/>
26. <https://www.diabetes.org.uk/our-research/about-our-research/our-impact/discovery-of-insulin>
27. <https://diabetes.org/blog/history-wonderful-thing-we-call-insulin>
28. https://en.wikipedia.org/wiki/Frederick_Banting
29. <https://www.youtube.com/watch?v=PfaQp5xN7fY>
30. <https://ccsmed.com/education/frederick-banting-and-charles-best-the-discovery-that-changed-diabetes-forever/>
31. <https://www.nobelprize.org/the-miracle-discovery-that-reversed-the-diabetes-death-sentence/>
32. <https://www.sciencedirect.com/science/article/pii/S2666970621000494>
33. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7569573/>
34. <https://www.who.int/europe/emergencies/situations/covid-19>

35. <https://www.cdc.gov/museum/timeline/covid19.html>
36. https://en.wikipedia.org/wiki/COVID-19_pandemic
37. <https://news.un.org/en/story/2023/05/1136367>
38. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10804921/>
39. <https://www.nm.org/healthbeat/medical-advances/new-therapies-and-drug-trials/covid-19-pandemic-timeline>
40. <https://www.npr.org/2025/03/11/nx-s1-5323221/5-years-ago-today-the-who-declared-covid-19-a-pandemic>
41. <https://www.facebook.com/7NewsAustralia/posts/it-has-been-six-years-the-world-health-organisation-who-declared-the-global-coronavirus-1393960459436989/>
42. <https://www.archives.gov/exhibits/influenza-epidemic/>
43. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7314216/>
44. https://archive.cdc.gov/www_cdc_gov/flu/pandemic-resources/1918-commemoration/1918-pandemic-history.htm
45. https://en.wikipedia.org/wiki/Spanish_flu
46. <https://familiesfightingflu.org/spanish-flu-deaths-comparing-pandemics/>
47. <https://www.paho.org/en/who-we-are/history-paho/purple-death-great-flu-1918>
48. <https://archives.upenn.edu/exhibits/penn-history/flu/>
49. <https://www.statnews.com/2021/09/20/covid-19-set-to-overtake-1918-spanish-flu-as-deadliest-disease-in-american-history/>
50. <https://www.census.gov/library/publications/1920/demo/1918-mortality-special-tables.html>
51. <https://www.youtube.com/watch?v=Gd6FOON8TIO>
52. <https://www.trumanlibraryinstitute.org/signing-of-the-medicare-and-medicaid-act-of-1965/>
53. <https://www.archives.gov/milestone-documents/medicare-and-medicaid-act>
54. https://www.senate.gov/artandhistory/history/minute/Medicare_Signed_Into_Law.htm
55. <https://www.facebook.com/RepFletcher/videos/on-this-day-in-1965-president-johnson-signed-medicare-and-medicaid-into-law-prog/1845235795970025/>
56. <https://healthlaw.org/announcement/medicare-and-medicaid-act-1965-2/>
57. <https://www.charlesriverhealth.org/medicare-and-medicaid-turn-60/>
58. <https://www.facebook.com/photo.php?fbid=1439892284591212&set=a.370884118158706&id=100057113636042>
59. https://en.wikipedia.org/wiki/Ten-percent-of-the-brain_myth
60. <https://mcgovern.mit.edu/2024/01/26/do-we-use-only-10-percent-of-our-brain/>
61. <https://faculty.washington.edu/chudler/tenper.html>
62. <https://www.psychologytoday.com/us/blog/common-sense-science/202505/beyond-the-myth-that-we-use-only-10-percent-of-our-brains>
63. <https://www.facebook.com/AllenInstitute/videos/-do-we-really-only-use-10-of-our-brain-and-do-we-get-special-superpowers-if-we-u/1609758340458608/>

64. <https://www.youtube.com/watch?v=e6wWw2B3Dz0>
65. <https://www.psychologicalscience.org/uncategorized/myth-we-only-use-10-of-our-brains.html>
66. <https://www.educationalneuroscience.org.uk/resources/neuromyth-or-neurofact/we-only-use-10-of-our-brains/>
67. <https://www.wired.com/2014/07/everything-you-need-to-know-about-the-10-brain-myth-explained-in-60-seconds/>
68. <https://www.britannica.com/story/do-we-really-use-only-10-percent-of-our-brain>
69. <https://study.com/learn/lesson/video/lock-key-model-vs-induced-fit-model.html>
70. <https://www.biologyonline.com/dictionary/lock-and-key-model>
71. <https://study.com/learn/lesson/lock-key-model-vs-induced-fit-model.html>
72. <https://synapse.patsnap.com/article/how-do-enzymes-work-lock-and-key-vs-induced-fit-model>
73. <https://www.slideshare.net/slideshow/lock-and-key-model-amp-induced-fit-model/250204228>
74. <https://www.youtube.com/watch?v=MHN9FLuJtXY>
75. <https://microbenotes.com/lock-and-key-model-mode-of-action-of-enzymes/>
76. <https://www.sciencedirect.com/topics/biochemistry-genetics-and-molecular-biology/lock-key-model>
77. <https://www.youtube.com/watch?v=EiMBsgNZh-M>
78. <https://en.wikipedia.org/wiki/Humorism>
79. <https://www.nlm.nih.gov/exhibition/shakespeare-and-the-four-humors/index.html>
80. <https://www.gavi.org/vaccineswork/funny-science-hippocratic-medicine-and-four-humours>
81. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3595629/>
82. https://stanfordbloodcenter.org/the_four_humour/
83. <https://www.youtube.com/watch?v=9k4RzFe6s8o>
84. <https://ubwp.buffalo.edu/history-of-cds/mesopotamia/ancient-theory-of-elements-and-humors/>
85. https://www.reddit.com/r/SeveranceAppleTVPlus/comments/tip55ku/the_four_temperaments_and_the_four_temperaments_humors/
86. https://www.reddit.com/r/EnglishLearning/comments/7oaigb/laziness_breeds_humors_of_the_blood/
87. <https://www.northwestcareercollege.edu/blog/the-four-humors-an-ancient-take-on-phlebotomy/>
88. <https://www.folger.edu/blogs/shakespeare-and-beyond/the-four-humors-eating-in-the-renaissance/>
89. <https://maryhrovat.com/humors-and-temperaments-blood-and-phlegm/>
90. <https://www.bbc.com/future/article/20250915-the-ancient-origins-of-todays-personality-types>
91. <https://becker.wustl.edu/news/humors-and-you/>
92. https://en.wikipedia.org/wiki/Four_temperaments
93. <https://www.bswhealth.com/blog/can-stress-cause-ulcers>
94. <https://health.clevelandclinic.org/can-stress-give-you-an-ulcer>
95. <https://www.muhealth.org/our-stories/what-causes-stomach-ulcers-dont-assume-spicy-foods-are-blame>
96. <https://northeastdigestive.com/blog/can-stress-cause-a-peptic-ulcer/>

97. <https://www.healthline.com/health/stomach-ulcer-diet>
98. <https://www.omicsonline.org/open-access-pdfs/exploring-the-connection-between-stress-diet-and-gastric-ulcers.pdf>
99. <https://4rai.com/2023/02/01/are-stomach-ulcers-caused-by-stress/>
100. <https://rockfordgi.com/stress-and-stomach-ulcers/>
101. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12286954/>
102. <https://www.carygastro.com/blog/understanding-stress-ulcers>
103. <https://ods.od.nih.gov/factsheets/VitaminC-HealthProfessional/>
104. <https://www.mayoclinic.org/healthy-lifestyle/nutrition-and-healthy-eating/expert-answers/vitamin-c/faq-20058030>
105. <https://www.goodrx.com/vitamin-c/dosage>
106. <https://www.webmd.com/diet/features/the-benefits-of-vitamin-c>
107. <https://ods.od.nih.gov/factsheets/VitaminC-Consumer/>
108. <https://www.nhs.uk/conditions/vitamins-and-minerals/vitamin-c/>
109. <https://www.health.harvard.edu/healthy-aging-and-longevity/by-the-way-doctor-whats-the-right-amount-of-vitamin-c-for-me>
110. <https://www.naturemade.com/blogs/health-articles/how-much-vitamin-c-should-you-take>
111. <https://www.uclahealth.org/news/article/vitamin-c-why-you-need-it-and-how-get-it>
112. <https://www.ncoa.org/article/vitamin-c-and-older-adults-how-much-is-enough/>
113. <https://medlineplus.gov/genetics/understanding/basics/howmanychromosomes/>
114. <https://www.genome.gov/about-genomics/fact-sheets/Chromosomes-Fact-Sheet>
115. <https://en.wikipedia.org/wiki/Chromosome>
116. <https://geneticsunzipped.com/transcripts/2019/4/25/the-case-of-the-missing-human-chromosomes>
117. <https://www.youtube.com/watch?v=Xgk7Zlwmau4>
118. <https://learn.genetics.utah.edu/content/basics/chromosomes/>
119. <https://www.nature.com/scitable/content/The-chromosome-number-in-humans-a-brief-15575/>
120. https://www.reddit.com/r/Biochemistry/comments/vy2pok/how_many_chromosomes_in_the_human_body/
121. <https://ubiehealth.com/doctors-note/how-many-chromosomes-do-humans-have-46-normal-37-q23e4>
122. <https://www.facebook.com/mplus.gov/videos/how-many-chromosomes-do-people-havein-humans-each-cell-normally-contains-23-pair/207809177852696/>
123. <https://my.clevelandclinic.org/health/body/chromosomes>
124. <https://www.choicedna.com/how-many-chromosomes-do-humans-have/>
125. <https://www.youtube.com/watch?v=Al7FueXTCKs>
126. <https://www.aaaai.org/tools-for-the-public/conditions-library/allergies/vaccine-myth-fact>
127. <https://www.cdc.gov/vaccines/basics/explaining-how-vaccines-work.html>
128. <https://www.who.int/health-topics/vaccines-and-immunization>

129. <https://www.hopkinsmedicine.org/health/treatment-tests-and-therapies/vaccines>
130. <https://www.cdc.gov/vaccines-children/diseases/index.html>
131. <https://www.sciencedirect.com/science/article/abs/pii/S0264410X23015062>
132. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7371956/>
133. <https://www.nhs.uk/vaccinations/why-vaccination-is-important-and-the-safest-way-to-protect-yourself/>
134. <https://publichealth.jhu.edu/2025/the-health-and-economic-benefits-of-vaccines>
135. <https://www.aaaai.org/tools-for-the-public/conditions-library/allergies/vaccine-myth-fact>
136. <https://www.cdc.gov/vaccines/basics/explaining-how-vaccines-work.html>
137. <https://www.cdc.gov/colorectal-cancer/screening/index.html>
138. <https://uspreventiveservicestaskforce.org/uspstf/recommendation/colorectal-cancer-screening>
139. <https://www.cancer.org/cancer/types/colon-rectal-cancer/detection-diagnosis-staging/acs-recommendations.html>
140. <https://blogs.cdc.gov/genomics/2018/08/30/when-should-you-be-screened/>
141. <https://colorectalcancer.org/screening-prevention/get-screened-45>
142. <https://pubmed.ncbi.nlm.nih.gov/34794816/>
143. https://www.cdc.gov/pcd/issues/2015/14_0533.htm
144. <https://www.uchicagomedicine.org/forefront/cancer-articles/2021/may/new-guidelines-lower-colorectal-screening-age-from-50-to-45>
145. <https://www.cologuardhcp.com/about/crc-screening-guidelines>
146. <https://clevelandadrc.org/education-resources/new-fda-approved-alzheimers-treatments/>
147. <https://www.nature.com/articles/d41573-024-00116-1>
148. <https://www.alz.org/alzheimers-dementia/treatments/lecanemab-leqembi>
149. <https://www.fda.gov/drugs/news-events-human-drugs/fda-approves-treatment-adults-alzheimers-disease>
150. <https://www.mayoclinic.org/diseases-conditions/alzheimers-disease/in-depth/alzheimers-treatments/art-20047780>
151. <https://www.brightfocus.org/news/fda-approves-at-home-injectable-leqembi-for-alzheimers/>
152. <https://www.alz.org/getmedia/4f4ca289-a2c6-4df9-8cdf-390365bd477e/alzheimers-dementia-fda-approved-treatments-for-alzheimers-ts.pdf>
153. <https://stanfordhealthcare.org/medical-conditions/brain-and-nerves/alzheimers-disease/treatments/medications.html>
154. <https://health.stonybrookmedicine.edu/healthcast-episode-6-new-fda-approved-infusion-drugs-for-early-stage-alzheimers/>
155. <https://www.nia.nih.gov/health/alzheimers-treatment/how-alzheimers-disease-treated>
156. <https://www.npr.org/2026/01/29/nx-s1-5689902/us-life-expectancy-rises>
157. <https://www.aha.org/news/headline/2026-01-30-cdc-us-life-expectancy-hits-all-time-high-79-years>
158. <https://www.cdc.gov/nchs/products/databriefs/db548.htm>

159. <https://www.scientificamerican.com/article/u-s-life-expectancy-hits-all-time-high/>
160. <https://www.optimistdaily.com/2026/02/us-life-expectancy-hits-all-time-high-and-itll-probably-keep-rising/>
161. <https://www.healthsystemtracker.org/chart-collection/u-s-life-expectancy-compare-countries/>
162. <https://www.cnn.com/2026/01/29/health/life-expectancy-mortality-rate-death>
163. <https://worldpopulationreview.com/state-rankings/life-expectancy-by-state>
164. <https://www.facebook.com/witntv/posts/us-life-expectancy-hit-an-all-time-high-in-2024-cdc-saysstory-httpswwwwitncom202/1309829581177080/>
165. <https://www.ssa.gov/oact/STATS/table4c6.html>
166. <https://www.worldometers.info/demographics/life-expectancy/>
167. <https://www.powershealth.org/about-us/newsroom/health-library/2026/01/30/us-life-expectancy-hits-record-high-as-death-rates-fall-nationwide>
168. <https://www.usnews.com/news/health-news/articles/2026-01-30/u-s-life-expectancy-hits-record-high-as-death-rates-fall-nationwide>
169. <https://www.healthpolicyohio.org/health-policy-news/2026/01/30/life-expectancy-in-us-reaches-record-high-as-overdose-deaths-decline>
170. <https://www.seniorliving.org/research/life-expectancy/>
171. <https://www.fda.gov/drugs/postmarket-drug-safety-information-patients-and-providers/fdas-concerns-unapproved-glp-1-drugs-used-weight-loss>
172. <https://www.goodrx.com/conditions/weight-loss/new-weight-loss-drugs>
173. <https://investor.lilly.com/news-releases/news-release-details/fda-approves-lillys-foundayotm-orforglipron-only-glp-1-pill>
174. <https://www.nature.com/articles/d41573-026-00059-9>
175. <https://www.fda.gov/drugs/drug-alerts-and-statements/fda-clarifies-policies-compounders-national-glp-1-supply-begins-stabilize>
176. <https://www.cnn.com/2026/04/01/eli-lilly-glp-1-pill-foundayo-approved-for-obesity.html>
177. <https://www.fda.gov/drugs/drug-safety-communications/update-fdas-ongoing-evaluation-reports-suicidal-thoughts-or-actions-patients-taking-certain-type>
178. <https://www.youtube.com/watch?v=HDyNTpBUYxc>
179. <https://jamanetwork.com/journals/jama/fullarticle/2844678>
180. <https://www.heart.org/en/health-topics/high-blood-pressure/the-facts-about-high-blood-pressure/high-blood-top-10>
181. <https://www.nhlbi.nih.gov/health/high-blood-pressure>
182. <https://www.ahajournals.org/doi/10.1161/CIR.0000000000001356>
183. <https://www.health.harvard.edu/heart-health/reading-the-new-blood-pressure-guidelines>
184. <https://utswwmed.org/medblog/updated-blood-pressure-guidelines/>
185. <https://www.yalemedicine.org/news/blood-pressure-goal-the-number-your-doctor-wants-to-see>

186. <https://news.cuanschutz.edu/news-stories/more-americans-meet-criteria-for-high-blood-pressure-under-new-guidelines>
187. <https://millionhearts.hhs.gov/data-reports/hypertension-prevalence.html>
188. <https://www.ahajournals.org/doi/10.1161/HYPERTENSIONAHA.125.25466>
189. <https://www.getreliefresponsibly.com/use-pain-medicine-safely/adult-acetaminophen-dosing>
190. <https://www.goodrx.com/tylenol/dosage>
191. <https://www.mayoclinic.org/drugs-supplements/acetaminophen-oral-route-rectal-route/description/drg-20068480>
192. <https://ubiehealth.com/doctors-note/acetaminophen-dosage-differences>
193. <https://www.tylenol.com/safety-dosing/dosage-for-adults>
194. <https://northeastpeds.com/resources/acetaminophen-tylenol-dosing-chart/>
195. <https://www.tylenolprofessional.com/products-dosage-adult>
196. https://www.guthrie.org/sites/default/files/versed_dosing_sheet.pdf
197. <https://www.cancer.org/cancer/risk-prevention/diet-physical-activity/body-weight-and-cancer-risk/body-mass-index-bmi-calculator.html>
198. <https://calculat.io/en/weight/bmi-imperial/5-ft-9-in--175-lbs>
199. <https://www.nhlbi.nih.gov/calculate-your-bmi>
200. <https://www.health.harvard.edu/bmi-calculator>
201. <https://www.verywellhealth.com/body-mass-index-bmi-5210240>
202. <https://www.nghs.com/bariatrics/bmi-calculator>
203. <https://www.premierhealth.com/premier-weight-loss-solutions/weight-loss-resources/bmi-calculator>
204. <https://www.ilbs.in/bmicalculator.html>
205. <https://www.dignityhealth.org/central-coast/locations/marianregional/services/weight-loss-surgery/calculate-your-bmi>
206. <https://www.cdc.gov/bmi/adult-calculator/index.html>
207. <https://www.mcw.edu/calculators/creatinine-clearance>
208. https://www.kidney.org/professionals/gfr_calculatorCoc
209. <https://www.hiv.uw.edu/page/clinical-calculators/crcl>
210. <https://clinicalc.com/kinetics/crcl.aspx>
211. <https://www.ncbi.nlm.nih.gov/books/NBK544228/>
212. <https://oncodaily.com/calculators/cockcroft-gault-creatinine-clearance-calculator>
213. <https://emedicine.medscape.com/article/2117892-overview>
214. <https://www.merckmanuals.com/professional/multimedia/clinical-calculator/creatinine-clearance-estimate-by-cockcroft-gault-equation-si-units>
215. <https://reference.medscape.com/calculator/51/crcl-cockcroft-gault>
216. <https://txwb.crab.org/TXWB/CreatinineClearanceCalculator.aspx>
217. <https://www.hepatitisb.uw.edu/page/clinical-calculators/crcl>

218. <https://handbook.ggcmedicines.org.uk/guidelines/infections/manual-calculation-of-creatinine-clearance/>

219. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8848562/>

220. <https://www.instagram.com/reel/DDexyGyNxxgD/>

221. <https://www.droracle.ai/articles/1072983/how-can-i-convert-amoxicillin-dosing-to-a-liquid>

222. <https://www.goodrx.com/amoxicillin/dosage>

223. <https://www.facebook.com/Fellownurses/posts/if-a-doctor-prescribes-250mg-of-amoxicillin-for-your-patient-and-pharmacy-suppli/1317805423679190/>

224. <https://healthify.nz/tools/a/amoxicillin-paediatric-dose-calculator>

225. <https://www.gauthmath.com/solution/1987269415485956/Amoxicillin-suspension-contains-250-mg-in-5-mL-The-patient-requires-500-mg-How-m>

226. <https://www.cdc.gov/diabetes/diabetes-testing/index.html>

227. <https://www.mayoclinic.org/diseases-conditions/diabetes/diagnosis-treatment/drc-20371451>

228. https://diabetesjournals.org/care/article/49/Supplement_1/S27/163926/2-Diagnosis-and-Classification-of-Diabetes

229. <https://www.dexcom.com/en-us/faqs/what-is-a-normal-fasting-blood-glucose-for-a-person-with-diabetes>

230. <https://diabetes.org/living-with-diabetes/treatment-care/checking-your-blood-sugar>

231. <https://www.aafp.org/pubs/afp/issues/1998/1015/p1287.html>

232. <https://www.nutrisense.io/blog/blood-sugar-level-charts>

233. <https://www.healthline.com/health/diabetes/blood-sugar-levels-chart-by-age>

234. <https://www.levels.com/blog/what-should-my-glucose-levels-be-ultimate-guide>

235. https://www.reddit.com/r/statistics/comments/81jepm/how_do_you_explain_in_layman_terms_why_p04_is_not/

236. <https://pmc.ncbi.nlm.nih.gov/articles/PMC6532382/>

237. <https://blog.minitab.com/en/blog/adventures-in-statistics-2/how-to-correctly-interpret-p-values>

238. <https://mathenchant.wordpress.com/2024/01/15/what-a-96-percent-chance-doesnt-mean/>

239. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4877414/>

240. <https://www.scribbr.com/statistics/confidence-interval/>

241. https://www.reddit.com/r/explainlikeimfive/comments/cn30wo/eli5_what_does_statistically_significant_mean/

242. <https://www.investopedia.com/terms/p/p-value.asp>

243. <https://www.facebook.com/groups/RxPrep.Study.Group/posts/1074566466086562/>

244. <https://data36.com/statistical-significance-in-ab-testing/>

245. <https://www.youtube.com/watch?v=vemZtEM63GY>

246. https://en.wikipedia.org/wiki/68-95-99.7_rule

247. <https://wmed.edu/sites/default/files/P-VALUES SIMPLIFIED.pdf>

248. <https://allnurses.com/what-tid-prn-really-mean-t601863/>

- 249. <https://www.ncbi.nlm.nih.gov/books/NBK546195/table/ch2.tab1/>
- 250. <https://www.mypcnow.org/fast-fact/prn-range-analgesic-orders/>
- 251. <https://rk.md/2014/difference-between-tid-q8h-prn-dosing/>
- 252. <https://surgery.utoronto.ca/sites/default/files/Pain Management-PRN Orders.pdf>
- 253. https://www.hospitalmedicine.org/globalassets/clinical-topics/clinical-pdf/pain_management_inpatient_order_set_ucsd_5_21_15.pdf
- 254. <https://one-program.org/wp-content/uploads/2020/03/NSAID-and-Opioid-Therapy-Table.pdf>
- 255. <https://pami.emergency.med.jax.ufl.edu/resources/dosing-guide/>